Network design practice 3-1

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1 Assignment 1

```
action set_direction(bit<1> dir) {
    direction = dir;                    /* Assignment1 : set direction of incoming(outgoing) packets */
}

/* Port checking table for setting direction of incoming/outgoing
table check_ports {
    key = {
        standard_metadata.ingress_port: exact;
        standard_metadata.egress_spec: exact;
    }
```

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Assigned direction to dir.

```
if (hdr.tcp.isValid()){
    if (check_ports.apply().hit) {
        compute_hashes(hdr.ipv4.srcAddr, hdr.ipv4.dstAddr, hdr.tcp.srcPort, hdr.tcp.dstPort);
    }
    else {
        compute_hashes(hdr.ipv4.dstAddr, hdr.ipv4.srcA                    Port);
    }
    // Packet comes from internal network
    if (direction == 0){
        bloom_filter_1.write(reg_pos_one, 1);       /* Assignment1 : write value of first register as 1 */
        bloom_filter_2.write(reg_pos_two , 1);      /* Assignment1 : write value of second register as 1 */
    }
    // Packet comes from outside
    else if (direction == 1){
        bloom_filter_1.read(reg_val_one , reg_pos_one );    /* Assignment1 : load value of first register to reg_va
        bloom_filter_2.read(reg_val_two , reg_pos_two );    /* Assignment1 : load value of second register to reg_v

        /* Assignment1 : write if statement to  incoming packets marked with register values of 0 - in order to dro
        if (reg_val_one == 0 || reg_val_two == 0) {
            drop();
        }
    }
```

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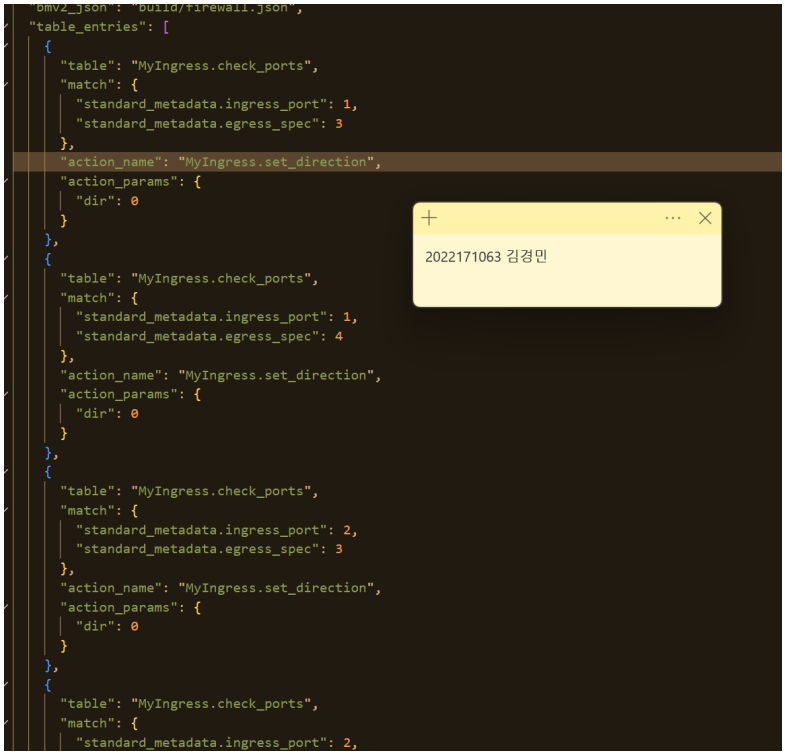
Using syntax, read register value and write using register position.



```
  "bmv2_json": "build/firewall.json",
  "table_entries": [
    {
      "table": "MyIngress.check_ports",
      "match": {
        "standard_metadata.ingress_port": 1,
        "standard_metadata.egress_spec": 3
      },
      "action_name": "MyIngress.set_direction",
      "action_params": {
        "dir": 0
      }
    },
    {
      "table": "MyIngress.check_ports",
      "match": {
        "standard_metadata.ingress_port": 1,
        "standard_metadata.egress_spec": 4
      },
      "action_name": "MyIngress.set_direction",
      "action_params": {
        "dir": 0
      }
    },
    {
      "table": "MyIngress.check_ports",
      "match": {
        "standard_metadata.ingress_port": 2,
        "standard_metadata.egress_spec": 3
      },
      "action_name": "MyIngress.set_direction",
      "action_params": {
        "dir": 0
      }
    },
    {
      "table": "MyIngress.check_ports",
      "match": {
        "standard_metadata.ingress_port": 2,
```

2022171063 김경민

```
"table_entries": [
  {
    "match": {
      "standard_metadata.ingress_port": 3,
      "standard_metadata.egress_spec": 1
    },
    "action_name": "MyIngress.set_direction",
    "action_params": {
      "dir": 1
    }
  },
  {
    "table": "MyIngress.check_ports",
    "match": {
      "standard_metadata.ingress_port": 4,
      "standard_metadata.egress_spec": 1
    },
    "action_name": "MyIngress.set_direction",
    "action_params": {
      "dir": 1
    }
  },
  {
    "table": "MyIngress.check_ports",
    "match": {
      "standard_metadata.ingress_port": 3,
      "standard_metadata.egress_spec": 2
    },
    "action_name": "MyIngress.set_direction",
    "action_params": {
      "dir": 1
    }
  },
  {
    "table": "MyIngress.check_ports",
    "match": {
      "standard_metadata.ingress_port": 4,
      "standard_metadata.egress_spec": 2
    },
    "action_name": "MyIngress.set_direction",
    "action_params": {
      "dir": 1
```

2022171063 김경민

In s1-runtime.json, replicate the table, and add protocol labeling packets coming in and out. All packets inbound are checked if dir=1. All packets outbound are checked if dir=0.

```
requests.txt

E0414 11:33:12.723200155   69306 fork_posix.cc:63]           Fork support is on
ly compatible with the epoll1 and poll polling strategies
mininet> iperf h1 h3
*** Iperf: testing TCP bandwidth between h1 and h3
*** Results: ['5.1 Mbits/sec', '6.4 Mbits/sec']
mininet> iperf h3 h1
*** Iperf: testing TCP bandwidth between h3 and h1
```

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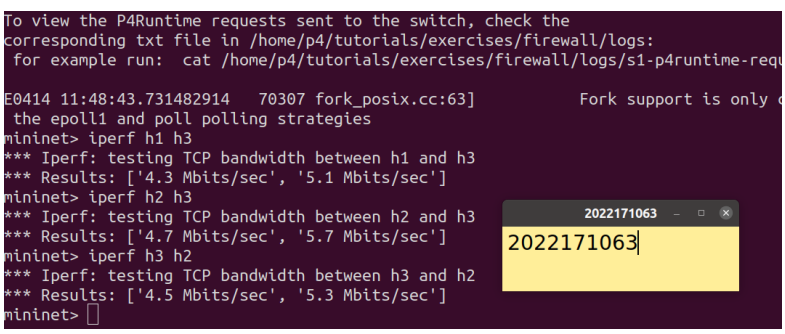
Confirmed ping from h1 to h3 is successful. Ping from h3 to h1 is blocked due to firewall.

2 Assignment 2

Figure 1: Modifying dir

Adjusted table content to check all packets bound for h2 if dir=0, to detour firewall.



Ping from h3 successfully reaches h2, and vice versa.

3 Assignment 3

To accomodate more than 10,000 flows without hash collision, the value of #define BLOOM_FILTER_ENTRIES (line 10 from firewall.p4) must be revised to at least 10,000.

The size of bloom filter entries must be larger than the total number of flows handled to ensure lower possibilities for hash collision to occur.

Since two switch registers are assigned to the each flow, the original resource usage was 2*4092=8192. Now to inhabit much more flows than that, the usage increases to 2*10,000=20,000.

The register size is defined at line 116 and 117 of firewall.p4.